A Citation-Calibrated Model of the Downstream Consequences of Worker Displacement

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October 2026
Parameter set v1.53

Abstract

We present `downstream`, an open-source model that propagates worker-displacement exposure through a ledger of published effects. It distinguishes earnings gaps, direct effects, relative risks, and jobs-per-job multipliers. Each parameter retains its citation, evidence precision tier, population scope, and uncertainty band. Scenario counts require verified baselines. Parameter sampling, exact moment calculations, sensitivity analysis, and named structural variants expose uncertainty in both estimates and composition choices. Current China-shock retrodictions show mixed results: family-status contrasts are covered, while mortality discrepancies and lower-quartile earnings underprediction remain. Experimental extensions compare household earnings trajectories, age-stratified mortality, and age-timed child earnings values, and propagate documented baseline and declared exposure uncertainty separately. Independent event back-tests remain blocked on data, and prospective forecasts have not yet been registered. We report those limits alongside the implemented methods. Outputs are conditional population-average estimates, not predictions about individuals or proof that an exposure was caused by a particular filing or employer. A current mathematical review identifies mortality transport, baseline, and place-counterfactual limits; numerical consistency alone does not establish causal validity.

1 Introduction

Worker displacement has consequences beyond a worker's immediate earnings. The underlying studies measure different outcomes in different populations: worker earnings, mortality, and children's later earnings are prominent examples [32, 53, 43]. A combined model must state how those estimates can be used together. A citation alone does not justify multiplication: a jobs-per-job multiplier and an intergenerational earnings elasticity act on different quantities.

This model makes those distinctions explicit. It provides a typed ledger, a versioned parameter set, reproducible uncertainty calculations, and validation results that retain misses. Its present implementation is a conditional propagation model with population aggregation, with a limited annual household earnings prototype. It does not simulate individual labor-market matching, migration, or behavioral interactions. The term microsimulation describes the intended family-level use; it should not imply that the code currently advances a representative synthetic population through an annual life cycle.

2 Relation to other modeling approaches

2.1 Life-course and household microsimulation

The intended person-and-place simulator belongs to an established research tradition. SimPaths projects individual and household histories across work, family relationships, health, and finances in an open-source framework [9]. It is a close architectural reference for dynamic feedback and household context. LifeSim combines developmental, economic, social, and health outcomes across an English birth cohort's lifespan

[49]. It is especially relevant to linking childhood circumstances with later outcomes. Neither project should be reduced to a collection of isolated outcome multipliers.

Urban Institute's DYNASIM ages individuals and families through demographic and economic transitions, including marriage, divorce, work, earnings, disability, and retirement [54]. The Future Elderly Model connects health and economic outcomes to evaluate alternative health and policy scenarios [55]. These programs provide useful precedents for baseline trajectories, family structure, scenario analysis, and empirical evaluation. Their domains and target populations differ; this comparison is selective, not a systematic ranking of model quality.

Downstream currently has a narrower implementation: cited response coefficients and count conversions conditional on an externally supplied exposure. Its proposed contribution is an inspectable connection between evidence applicability, composition restrictions, uncertainty, and accessible scenario comparison. Openness, simulation of multiple domains, and citation of evidence alone do not establish novelty. Whether its composition rules improve out-of-sample performance remains a research question, not a demonstrated advantage over these predecessors.

2.2 Causal composition and population consequences

Our design separates three tasks often combined in larger models: estimating an exposure, propagating its consequences, and evaluating the result. The exposure is supplied by the caller. The household prototype reuses observed employment transitions under declared assumptions; a full dynamic population model would additionally need entry, exit, and linked demographic and economic transitions. An interacting-agent model would need behavioral rules and a network; neither is present here.

Regional multipliers enter as a separate count boundary, following the local-employment literature [42]. They are not passed through an earnings chain. Evidence synthesis considers published individual studies and meta-analyses, but numerical rows with unresolved source support are withdrawn from calculation. We compose selected links under declared assumptions rather than pool unlike estimands into a single effect. The model is therefore an auditable synthesis of conditional responses, not a newly identified causal estimate of the entire system.

3 Model and evidence

Figure 1 separates supplied inputs, model operations, and reported outputs. Its arrows describe computation; they do not establish that a causal pathway is identified.

3.1 Exposure and outcomes

A scenario supplies displaced-worker exposure N, children per exposed worker C, independently documented net tradable-job loss T, a job class, and a mortality window Y. The tradable-share field is metadata only; it does not infer T from N. The worker module reports earnings and mortality responses. The child module reports earnings across three descendant generations. Family and community modules expose additional pathways, but their availability in a module does not imply that the scenario count interface reports them. The current count interface returns local service jobs, excess deaths, and child lifetime earnings losses, subject to verified baselines and outcome-specific applicability checks. Default results are illustrative reference calculations; a documented target population and reviewed transport decision are required for headline eligibility. The mortality count is a reference scenario for US men aged 45–54. Its displacement effect comes from a study of high-seniority men. The child earnings link comes from Canadian father–son firm closures and uses a US male lifetime-earnings baseline in 2024 dollars. Applying either result to a mixed or unknown population requires a separate transport argument. Receiving-community crime links use source-specific population shares: an undocumented-share percentage-point change for Light–Miller's log-rate response, and a regular-resident share ratio for Bianchi's log-log response. They are separate evidence inputs, not consequences inferred from displaced-worker exposure. The named adapter retains different source windows and fixed-effects versus IV roles. Bianchi's calculation uses the available 2010 working-paper version; reconciliation with the 2012 published article remains pending. Correct units do not

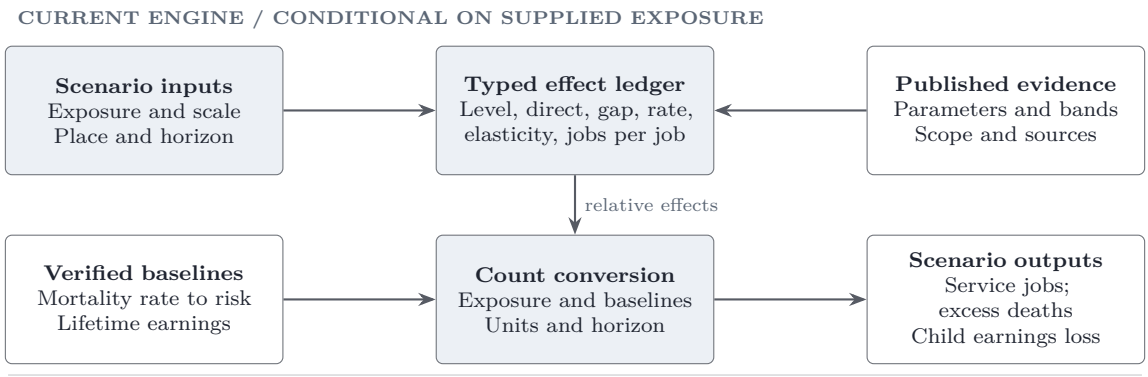


Figure 1: The current calculation boundary. Exposure is supplied, not inferred. The ledger separates effect types before count conversion. A citation records provenance; it does not by itself establish transportability or causal composition.

establish target transport. Optional incident differences use the same supplied post population in both rate scenarios and concern an annual endpoint, not cumulative crimes. For supplied reference and post shares s_0, s_1, the rate ratios are $R = \exp\{100\beta(s_1 - s_0)\}$ for the unlogged percentage-point predictor [35], and $R = \exp\{\beta \log(s_1/s_0)\}$ for log shares [4]. The second expression requires positive shares. Resident-count inputs first form each share using its own population denominator, rather than treating immigrant-stock growth as share growth. For an annual baseline rate b per 100,000 and a supplied post population P_1, the signed endpoint difference is $(bP_1/100000)(R-1)$. Interpreting exponentiated log-rate predictions as arithmetic rate ratios assumes the residual retransformation factor is unchanged between scenarios. Neither this assumption nor compatible units establish target transport.

The entity adapter accepts either a documented exposure or a warehouse aggregate with a stored total and components. The warehouse mapping checks agreement between the headline and breakdown totals. It checks that the worker-equivalent base, wage-depression, and applicable WARN components reconcile within the upstream rounding tolerance. It preserves the total; it does not re-estimate a wage-gap formula or add a notice count to a worker count. Source and method references accompany the result. Person subjects use the same population-average impact framing as employers. This interface does not establish a causal link between filings and worker displacement.

3.2 Typed composition

Let v be a counterfactual-relative earnings level, with no loss represented by $v = 1$. Let p be a parameter and t an intergenerational transmission coefficient. The ledger distinguishes

$$\text{level:} \quad v' = vp, \tag{1}$$

$$\text{direct:} \quad v' = p, \tag{2}$$

$$\text{gap:} \quad v' = 1 - t(1 - v). \tag{3}$$

A rate step records a relative effect for later conversion against a baseline. A level ratio multiplies a count outside the chain. An elasticity converts an input percentage change into an outcome percentage change. These distinctions prevent a transmission coefficient from being mistaken for a relative earnings level.

For the current parameter points, the direct child level is 0.9076 and the grandchild level is 0.9492. Great-grandchild earnings are unavailable: the prior repeated-transmission row has unresolved numerical/source support and is withdrawn. These values come from the same code used by the CLI; the first is the direct child estimate and the grandchild value propagates its gap [43, 51, 20, 16]. The father's earnings shock is not multiplied into the directly estimated child effect. Repeated transmission beyond the identified generations remains a modeling assumption.

Even the first transmission requires an extra causal assumption: a between-family income association need not equal the response to a displacement intervention. If t is instead interpreted literally as a constant log–log elasticity, the finite-change mapping is $v' = v^t$. The implemented gap rule is its first-order expansion about $v = 1$, $v^t = 1 + t(v-1) + O((v-1)^2)$. Thus it is an explicit linear approximation, not an exact conversion of a log elasticity. The implemented `log_elasticity_transmission` ensemble variant compares the finite-change mapping with the default gap rule using the same cited coefficient. Both retain the structural-transmission label; neither is an independently identified transmission of a displacement effect.

Parallel divorce and family-size pathways are not silently added to the displacement pathway. The same affected children may appear in several studies, and summation can double-count their losses. A named structural variant exposes gap-additive composition as an experiment rather than making it the default.

3.3 Parameter custody

This build contains 51 available parameter rows in set v1.53. Sixteen historical records are withdrawn from calculation: seven source/estimand/unit mismatches and nine unverified numerical extractions. Their original values remain in an explicitly separate export archive with the source-review reasons; overrides and sampling cannot restore them. Appendix B includes only available rows generated directly from the parameter file. Precision tiers describe how the number was extracted: `EXACT` denotes a pinned table or full-text estimate; abstract/results variants identify the narrower source read; `canonical` marks an established value with remaining extraction work. Precision of extraction is distinct from strength of causal identification. Associational conversion factors and parallel evidence must retain their limitations even when transcribed exactly.

Each row also stores population scope and notes on its band. Bands can represent reported uncertainty, a range across specifications or studies, or an explicitly declared approximation. They are not interchangeable confidence intervals. The audit checks citation resolution, units, and parameter structure. It cannot prove that a source has been interpreted correctly. Full notes and evidence classes remain in the CSV and bibliography rather than being replaced by the compact appendix.

3.4 Count conversion

With annual mortality risk $m = 1 - e^{-\lambda}$ from baseline rate λ, source-offset odds ratios r_j, and lifetime earnings baseline L, the current scenario implements

$$q(r) = \frac{rm}{1 - m + rm}, \tag{4}$$

$$D = N\left\{(1-m)^Y - \prod_{j\in\mathcal{P}}[1 - q(r_j)]^{d_j}\right\}, \tag{5}$$

$$E = NC(1 - g_c)L, \tag{6}$$

$$J = T\ell_{\text{class}}, \tag{7}$$

where $\mathcal{P} = \{0, 1, 2{:}3, 4{:}5, 6{+}\}$ indexes the displacement-year and source-offset phases, and their durations are $d_0 = \min(Y, 1)$, $d_1 = \min\{\max(Y-1, 0), 1\}$, $d_{2:3} = \min\{\max(Y-2, 0), 2\}$, $d_{4:5} = \min\{\max(Y-4, 0), 2\}$, and $d_{6+} = \max(Y-6, 0)$. Here g_c is the child earnings level and ℓ is the service-job multiplier. Engine 0.2 converts the stored odds ratios to risks and accounts for survival depletion. The follow-up includes the displacement year as year 1. The default `source_profile` uses all five source-offset phases. The source working paper's Table 5[1] reports the displacement year, offsets +1, +2–3, and +4–5, and +6 and later. Under this convention, the +6 coefficient begins in follow-up year 7. The historical `source_aligned` option preserves the former incomplete three-phase approximation for reproducibility only. The optional `immediate_sustained` timing applies the +6 coefficient after the displacement year as a sensitivity assumption. The historical additive formula remains available as `legacy_additive`. Current V1 and scoreable V2 mortality calculations use the production survival kernel and the source-offset profile by default.

[1]Sullivan and von Wachter, NBER Working Paper 13626, PDF page 54: https://www.nber.org/system/files/working_papers/w13626/w13626.pdf.

The implementation evaluates small survival differences with log-survival ratios and `expm1`, preserving signed near-null effects instead of subtracting nearly equal probabilities. Parameter revision v1.51 re-executes the five Table 5 coefficient/standard-error transformations consistently at four decimals. Working-paper numerical extractions retain canonical publication tier pending published-version reconciliation; a precise transcription is not evidence of peer-reviewed version equivalence.

Default source-profile scenarios also report a persistence sensitivity that retains the early phases but sets the offset +6 onward odds ratio to one, first changing follow-up year 7. Both alternatives use the same resolved baseline, exposure, horizon, and early odds ratios. For the national reference of 1,000 workers and 20 years, the source profile gives 28.76 expected excess deaths and the no-sustained-excess alternative gives 20.52; their point difference is 8.24. Monte Carlo reports a parameter-only interval within each structure and calculates their difference within each shared parameter draw. The range of their points is a structural comparison, not a confidence interval or a probability-weighted model average. The alternative is motivated by the long-run divergence recorded for Eliason and Storrie [24]. It is a hybrid assumption, not a reproduction of that Swedish study or a substitution of hazard ratios for odds ratios. A nonsignificant estimate does not establish an exactly zero effect, and persistence beyond that study's follow-up remains an assumption.

Only verified baseline values enter a count. A missing baseline produces a named blocked outcome, or raises an exception in strict mode. The baseline file records source, population, units, and derivation. Table 1 lists its current values; some support modules beyond the count interface.

Table 1: Stored baselines. Units and source qualifications are discussed below.

Baseline	Value	Population
all cause mortality annual	0.004944	US prime-age men (45-54)
divorce 5y cumulative	0.1045	US married couples 5y window
ipv annual incidence	0.0027	US residents age 12 or older
youth crime participation	0.055948	US young men 18-24
median male lifetime earnings	2,591,418	US male median lifetime earnings

Mortality converts the CDC WONDER deaths-per-person-year rate to the annual probability m using the constant-hazard mapping. A person-time rate is not itself a cohort probability [13]; under a constant hazard λ, that probability is $1 - e^{-\lambda}$. The default scenario applies this conversion for a compatible mortality rate and holds its baseline age rate fixed. A separate experimental cohort projection advances age-band baselines as described below. Divorce uses the Census SIPP five-year marriage-survival complement, which is not a pure divorce incidence measure. The NCVS IPV rate counts incidents rather than unique victims. The youth-crime baseline counts arrests per young man rather than unique offenders. Lifetime earnings are in 2024 dollars, derived from SSA age-band earnings and BLS inflation data; summing cross-sectional medians approximates, rather than equals, the median lifetime total. Exact source tables, windows, and arithmetic are retained in `params/baselines.csv`.

3.4.1 Additive coefficients with unknown covariance

A separate diagnostic examines additive log-odds coefficients in the available published-version mortality manuscript [53]. It does not replace the production working-paper profile. The standard error of a phase sum cannot be obtained by adding variances unless the covariances are known or independence is justified. For one linear contrast $T = \sum_i a_i \beta_i$ with marginal errors σ_i, let $l_i = |a_i|\sigma_i$. Over positive-semidefinite covariance matrices with that diagonal, the sharp bounds are

$$\max\{0,\, 2\max_i l_i - \sum_i l_i\} \;\le\; \mathrm{SE}(T) \;\le\; \sum_i l_i. \tag{8}$$

To see attainability, represent the covariance as a Gram matrix of vectors with lengths σ_i. Align the weighted vectors for the upper bound; oppose the longest with the remainder, or close a polygon, for the lower bound. Signed weights are absorbed into vector direction. The extrema concern one contrast; separate

phases need not attain their extrema jointly. The implementation also reports the point plus or minus 1.96 times the upper bound as an outer normal-approximation envelope, conditional on supplied marginal errors. It is not a source-reported confidence interval, a simultaneous band, or a joint draw model. These bounds neither resolve source timing and version differences nor propagate nonlinear twenty-year mortality uncertainty.

3.5 Child applicability and age-timed valuation

The direct child source concerns Canadian sons aged 10–14 at paternal firm closure. A supplied child profile with a known incompatible age or sex blocks the child-dollar count; an unknown profile remains illustrative. Demographic matching alone is insufficient for reviewed use: the interface requires target-population and transport decisions, and an explicit income rank definition. Optional father-income-quartile contrasts use the source's Table 8 log-earnings coefficients and standard errors [43]. Quartiles refer to average paternal income in 1978–1979 and need a justified mapping to the target. Signed coefficients and their uncertainty are retained, including null-compatible and beneficial estimates. No response curve across child ages is inferred.

The original count E is an undiscounted synthetic career loss in 2024 dollars. Its baseline must be finite, nonnegative, and denominated in 2024 US dollars. Custom parameter directories also supply the subgroup study library, whose selected scale, population, and outcome window must match the consuming calculation. With known child ages, a supplied real discount rate r, and an event/valuation year, the optional valuation report instead computes

$$E_{\mathrm{PV}} = N(1-g_c)\sum_{i=1}^{C}\sum_{a\in A_i}\frac{L_a}{(1+r)^{a-a_i}}, \tag{9}$$

where a_i is child i's age at displacement and L_a is the synthetic annual earnings profile normalized to the stored career baseline. The report separates the observed adult-age window 25–33 from a constant-gap career projection over ages 20–64; ages before the event are excluded. Both impose a constant relative gap over their listed ages; neither recovers an identified annual causal path. Zero discounting of the career projection reproduces the original count for source-eligible ages. Sampled valuations add separate parameter-only intervals, holding child ages, discount rate, earnings profile, and survival/employment treatment fixed. These exploratory present values do not qualify for public headlines.

3.6 Experimental age-stratum mortality

The separate `mortality-cohort` command aggregates expected effects over explicitly supplied age/sex/tenure strata. Its current response support is men aged 30–69 at mass layoff or firm closure with at least six years of tenure. It uses age-group annual-logit contrasts from Sullivan and von Wachter's working-paper Table 7, held constant through follow-up under an explicit constant source-window log-odds projection assumption. These source-window contrasts are distinct from the default Table 5 phase profile. Source age selects the response at displacement; the response itself does not change as survivors age.

For each stratum the command compares fixed baselines with advancing age-band US all-cause rates for 2015–2019. Survival products are computed within strata before exact weighted aggregation. These are piecewise period rates, not single-age hazards or a forecast of future mortality trends. Unsupported strata remain unknown: if any positive exposure lacks a supported response, no whole-cohort total is reported. The reported bands are Cartesian envelopes of marginal coefficient ranges, not simultaneous confidence bands or cohort confidence intervals. Marginal ranges do not establish joint coverage; baseline variance, exposure error, cross-coefficient covariance, and transport error remain excluded. The projection is experimental and ineligible for public headlines.

3.7 Place resolution

Where measured place values exist in compatible units, baseline pooling uses $b_* = (1-w)b_{national} + wb_{local}$ with $w = n/(n+k)$. The precision count n comes from the input data; the prior precision k is a declared modeling choice. Missing local information keeps the national baseline and reports the reason.

The mobility layer derives a multiplier $M = 1 + 18\gamma(q - q_0)/100$, where q is the county's permanent-resident income-rank percentile and q_0 is the national reference. The exposure coefficient γ comes from Chetty and Hendren [15]. Applying an income-rank exposure effect as a modifier of a displacement loss is an additional assumption, not the estimand of that study. The count interface uses the same-place contrast $M(1 - g)$, equivalently a child multiplier $1 - M(1 - g)$, so a null displacement effect ($g = 1$) remains null at every place. The child modifier is applied once before transmission. Descendants inherit that adjusted gap. The historical repeated application is available as an explicit structural alternative, not the default. Place-aware sampling recomputes M for each parameter draw, while county measurements and pooling weights remain fixed.

This null-safe formulation does not identify a displacement-by-place interaction. If M changes the earnings baseline in both exposure arms, their level difference would be $M(1 - g)$; if it changes susceptibility to displacement, an independently supported effect modifier is needed. Place-specific child dollar losses remain exploratory and the national child-dollar result is the reference.

3.8 Experimental household comparison

The `household` command compares annual employment-state probabilities and gross household earnings over one to five years. Adults have supplied sex, age, starting employment state, and annual earnings if employed. Observed SIPP December-to-December 2021–2022 conditional transitions over employed, unemployed, and not-in-labor-force states are reused under explicit stationarity and Markov assumptions. Adults remain within ages 25–54 during follow-up. The event starts from an employed reference member and forces unemployment for a caller-supplied integer duration; any reemployment earnings multipliers require a separate declared assumption. Unemployment-origin recovery is observational and does not identify a displacement recovery response.

The implementation advances state probabilities exactly, sums member earnings expectations, and ages adults and children. Sparse transition cells require explicit acceptance. It does not estimate partner responses or joint household dependence, taxes, transfers, savings, marriage, fertility, or protective interventions. Children age without an inferred educational or earnings consequence. Household resources here mean gross earnings only. Survey-design uncertainty and sparse cells preclude a full household prediction interval; this prototype remains ineligible for public headlines.

4 Uncertainty and experiments

Figure 2 distinguishes the uncertainty objects used below. Each answers a different question.

4.1 Parameter sampling

Latin hypercube sampling stratifies each parameter marginal [40]. Positive ratio units default to log-uniform sampling; other undeclared marginals default to uniform bands. Rows can explicitly select normal or lognormal shapes. The current normal-family sampler clamps draws at the stated band limits. This is a bounded approximation with endpoint mass, not sampling from a renormalized truncated-normal density. The point is retained as the declared center when applicable.

Declared rank correlations can be imposed through an Iman–Conover reordering [30]. The shipped table contains 2 declared pairs linking worker earnings with peak and sustained mortality. Their direction is motivated by the shared recession context, but their moderate negative magnitude is a declared assumption, not an extracted sampling covariance. Unrolled copies of one intergenerational relationship share exactly one uncertainty draw, rather than independent copies. Sensitivity attribution groups those copies; independent-step analytic inference refuses chains that reuse the same relationship. Monte Carlo applies these pairs by default; analytic chains and standard Sobol experiments retain their independence assumptions. Each Monte Carlo output records the seed, draw count, sampler, and parameter version. Sample quantiles describe uncertainty conditional on the specified distributions; they do not include all model error.

Scenario output separates three objects: endpoint support envelopes, central 90% parameter-only intervals, and simulated realized mortality counts. For an integer cohort, let $S_0 = (1 - m)^Y$ and let S_1 be

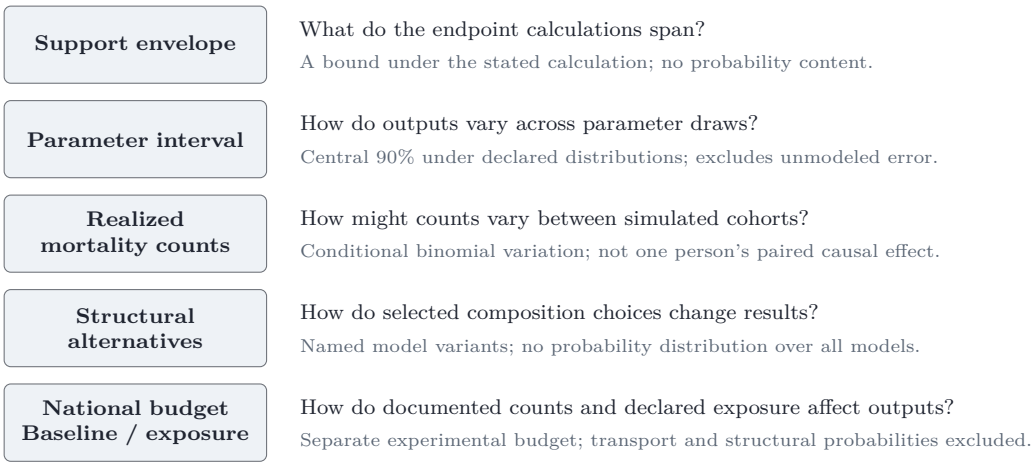


Figure 2: Five distinct uncertainty summaries. They are not interchangeable intervals. Default parameter draws hold baseline and exposure inputs fixed; the separate national budget relaxes those assumptions. Study transport and missing mechanisms remain outside both calculations.

the exposed survival product above. Conditional on each parameter draw,

$$K_0 \sim \text{Binomial}(N, 1 - S_0), \qquad K_1 \sim \text{Binomial}(N, 1 - S_1), \qquad K_0 \perp K_1. \tag{10}$$

Then $E[K_1 - K_0 \mid \theta] = N(S_0 - S_1)$. Both cohorts use the same resolved county or national baseline. Their independent difference is a reference-cohort contrast, not a paired individual causal effect. Fractional worker equivalents and the legacy additive mortality method do not receive this count simulation. Although the API calls it "posterior-predictive," heterogeneous elicited parameter bands are not a fitted joint Bayesian posterior; this is conditional predictive simulation under the declared input distributions.

4.2 Separate baseline, exposure, and process budget

The default scenario intervals above hold baseline and exposure inputs fixed. The separate experimental `uncertainty` command relaxes those restrictions for the national source-profile mortality scenario. Documented baseline deaths D_b and person-time T_b must match the verified rate and its sex, age, cause, calendar window, and geography. With a Poisson likelihood and a declared Jeffreys rate prior, it samples

$$\lambda \mid D_b, T_b \sim \text{Gamma}(D_b + \tfrac{1}{2},\ \text{rate} = T_b). \tag{11}$$

Exposure is drawn from a caller-declared discrete uniform distribution over integer bounds that bracket the reference cohort. That distribution is a sensitivity assumption, not an inferred confidence distribution. The calculation combines these inputs with parameter draws and retains the original fixed-input reference. Expected-effect intervals and independent-cohort binomial process contrasts are reported separately.

Baseline, exposure, and study parameters are assumed mutually independent, while the two mortality persistence structures share each draw. Transport error, probabilities over structures, missing mechanisms, age/time trends, and uncertainty in child structure and earnings profiles remain excluded. An optional, explicitly declared sensitivity layer multiplies each sampled phase's log odds ratio by a fixed transport scale. Separate cases retain the independent input pairing or align/oppose sampled exposure and baseline ranks with the signed source excess-death probability at the pinned baseline. Study vectors and auxiliary marginal samples are preserved; these selected rank arrangements are not bounds over all dependence structures. Cases retain both persistence structures and report paired differences from the independent source reference, without assigning case probabilities, pooling intervals, or claiming empirically calibrated transport error. Known population refusals remain in force. This national budget does not sample county posterior uncertainty or support demographic-profile overrides. It is not a complete uncertainty interval for an employer or individual and remains experimental.

4.3 Analytic moments and coverage

For an independent level multiplier X, raw moments obey $E[(VX)^k] = E[V^k]E[X^k]$. A direct step replaces the moments. For a gap step with independent T and V,

$$E[V'] = 1 - E[T] + E[T]E[V], \tag{12}$$

$$E[(V')^2] = 1 - 2E[T] + E[T^2] + 2(E[T] - E[T^2])E[V] + E[T^2]E[V^2]. \tag{13}$$

The implementation also propagates third moments for supported marginals. It uses them for skewness and Cornish–Fisher quantile approximations. Exact moments do not make those approximate quantiles exact. Unsupported distribution cases are refused. The identities also require that each new multiplier be independent of the accumulated quantity; reuse of a parameter can violate that condition.

Monte Carlo comparisons check the analytic calculations. Coverage closure experiments draw synthetic truths under the declared model and measure whether independently built intervals cover them. These tests assess interval machinery under its own assumptions. They cannot establish coverage for real-world events. Correlation stress experiments similarly price a hypothetical dependence choice without treating it as evidence.

4.4 Sensitivity, sweeps, and structural variants

Public scenario counts also attach deterministic effect-removal and one-input endpoint refits. Each experiment changes one admitted input while retaining the selected sources, baseline, population and other assumptions. A neutral ratio is one; a neutral additive/job or place-modifier coefficient is zero where defined by that outcome's implementation. The reported removal changes need not sum to the total because survival and composition are nonlinear. Removing an effect can also leave its observed support, so the experiment is a declared hypothetical comparison, not an identified causal share. Endpoint spans are conditional response ranges rather than variance indices. Fixed exposure, baseline and transport assumptions acquire no uncertainty distribution through these experiments.

The sensitivity interface estimates first-order and total Sobol indices using a Saltelli-style design [50, 46]. Seed replicates report design variation; that spread is not a confidence interval on the true index. Standard variance attribution assumes independent inputs. A place adjustment can make the mobility coefficient contribute to child-line sensitivity even though it has no effect at the national reference.

A separate correlated-block option groups all connected dependent parameters and swaps each joint block between independently seeded designs. It estimates attribution to independent blocks, not to individual dependent coefficients. Reusing a fixed copula rank template across the two designs invalidates that experiment; the current repair tests it against an additive model with known block variance shares.

A sweep pins one parameter inside its cited band and recomputes scenario counts. A value-of-information experiment narrows each parameter band around its point and compares output-band widths on a common seed. This is a precision-priority heuristic, not an expected utility calculation or a prediction of what a new study will find. Experiments carry a version marker and do not overwrite the published parameter set. Structural ensembles report named alternatives for gap additivity, transmission decay, finite-change log elasticity, and closure selection. Scenario mortality additionally reports the persistence comparison above. Their spread represents selected assumptions, not a probability distribution over all possible models.

The policy comparator subtracts caller-supplied exposure scenarios. It reports a conservative difference of support envelopes, with exact cancellation for identical cases. Shared-draw paired contrasts are a future improvement; the current envelope is not a probabilistic policy effect interval and does not identify a policy-to-exposure bridge.

4.5 Evidence synthesis and county likelihoods

The optional synthesis module implements DerSimonian–Laird heterogeneity estimation and inverse-variance random-effects pooling only after exact link, scale, population, and horizon checks, with declared treatment, comparison, outcome, and overlap metadata retained for review. Its intervals use a plug-in normal approximation; uncertainty in the estimated between-study variance is not propagated. Few-study

prediction intervals can be unreliable [18]. Matching scale strings alone does not enforce compatible follow-up, population, design, or independent samples. Metadata review remains necessary, and these outputs are not automatically admitted as parameters.

The standalone county module uses $D \mid \lambda \sim \text{Poisson}(T\lambda)$ with a gamma prior, where T is compatible person-time, and reports the constant-hazard conversion $1 - \exp(-\lambda)$ separately from the rate likelihood. It requires national-prior outcome, population, and window metadata to match the county observation. The county module is not integrated into scenario baseline uncertainty.

5 Validation results

5.1 V0: internal consistency

The direct child estimate is 0.9076 [0.844, 0.976], while the worker-shock path composed with intergenerational transmission is 0.89 [0.85, 0.94]. Their bands overlap. This checks compatibility between two modeled paths; it does not provide an independent validation of their causal claims.

5.2 V1: China-shock unit scorecard

Figure 3 plots the unit comparisons. Exact values are retained in Appendix A. The unit scorecard uses the exposure bridge and measured coefficients transcribed from Autor, Dorn, and Hanson [2]. The bridge maps an import shock into manufacturing displacement. The downstream mortality path uses a separate study [53], but comparing it with a commuting-zone male–female mortality differential requires strong population and outcome-transfer assumptions.

Measured points lie inside the modeled bands for 0 of 3 mortality variants; 3 of 3 modeled points exceed their measured targets. The pooled-adult prediction is 18.78 [5.51, 42.87] against 4.27 deaths per 100,000 adults per unit shock. A measured confidence interval and a modeled parameter band answer different questions; overlap is not proof of validity. The full JSON scorecard also reports whether the modeled point lies inside the measured interval. Age matching does not remove the distinction between an absolute mortality level and a sex differential, nor the transfer from high-seniority displaced workers to a broader exposed population.

The divorce path uses independent displacement studies [45, 14] and a marriage baseline. Its point estimates undershoot the observed family-status stock change at both windows. That stock includes widowhood and separation; divorce incidence alone cannot reproduce all its margins. Remarriage is a candidate missing mechanism, not an extracted explanation. Several additional target outcomes are withheld because the implemented evidence path would reuse the target coefficient rather than provide an independent comparison.

5.3 V1: commuting-zone contrasts and scoring

Figure 4 plots the panel contrasts on separate axes for family status and earnings. The deposited panel extract contains 1444 commuting-zone-period rows. Comparisons demean exposure and outcomes within period before contrasting high- and low-shock terciles. Pooling without that adjustment confounds period differences and changes the sign of the family-status comparison. The tests retain this failure mode.

The family-status contrasts are covered. Direct worker earnings losses alone substantially underpredict the observed lower-quartile earnings change. Adding the non-displaced wage response [3] reduces that gap but does not eliminate it. The earlier direct-only miss remains beside the composed result. The spillover extension was motivated by the observed miss; it is a diagnostic revision, not a fresh held-out success. Shared China-shock evidence further limits independence.

The model median slope is 0.079 against an observed slope of 0.28. CRPS is 0.1779 in percentage-point units; PIT is 1. CRPS summarizes distance between a predictive distribution and an observed point; PIT locates that point within the distribution [28]. Scoring against a published coefficient treats its point estimate as the target and does not integrate its measurement uncertainty. The upper-tail PIT and earnings residual show why selected tercile coverage should not be described as overall model validation.

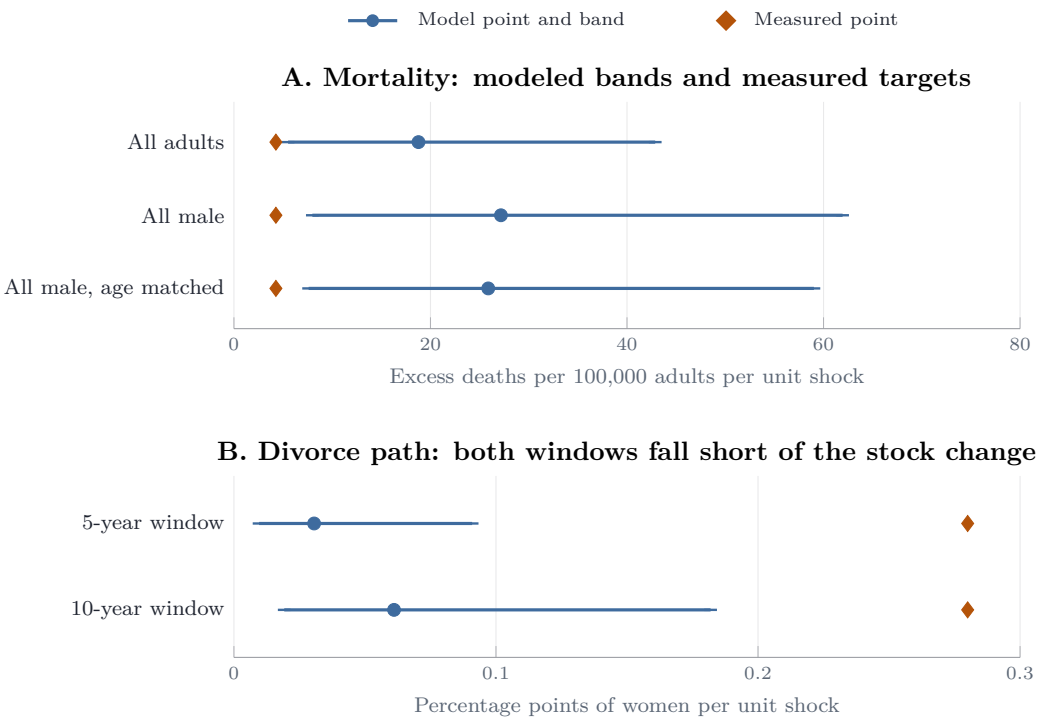


Figure 3: China-shock unit checks generated from the current engine. Blue lines are model bands, not a common confidence level; amber diamonds are measured point targets without their sampling intervals. Panel A compares modeled mortality with a male–female differential; panel B compares divorce with a broader family-status stock change. These target mismatches limit both comparisons. Numerical values appear in Appendix A.

5.4 V2 and V3: incomplete validation

Independent-event protocols exist for NAFTA, the auto crisis, and BRAC. Their scorecards are blocked on event-specific exposure bridges or measured outcome inputs. A tariff change cannot inherit an import-penetration displacement bridge merely because both concern trade. Descriptive employment loss does not supply a quasi-experimental measured outcome. No completed V2 event score is claimed here.

Prospective registration and scoring checks are implemented, but no real event has yet been registered. It requires forecasts fixed before outcomes are available, a declared exposure and time horizon, and a scoring rule fixed before evaluation. No prospective result is reported. Unit and panel V1 results are the current external checks.

5.5 Descriptive cohort and baseline checks

The reproducible CPS January-2024 reference describes known-year 2021–2023 displacements with lost-job tenure of at least three years, with a separate six-year-tenure subset. Survey weights reproduce the rounded published BLS total. Age and household composition are measured at interview and cannot silently stand in for historical event-age characteristics. Neither cohort identifies H-1B-specific displacement. The held-out SIPP checks compare baseline transition probabilities; they do not validate displacement effects or household recovery. The archived October 14 expanded evaluation reports a cohort reproduction difference of 128.318 workers against the published 2,578,000-worker benchmark, within its 1,000-worker rounding tolerance. The SIPP check includes 54 held-out calibration cells; the largest absolute transition-probability difference is 0.3759. The study library produces 0 compatible pooled reports and 23 explicit pooling refusals. These archived results are retained in `validation/followthrough_evaluation_expanded_20261014.json`. Its compatible-pooling refusals preserve population, horizon, scale, and overlap differences rather than

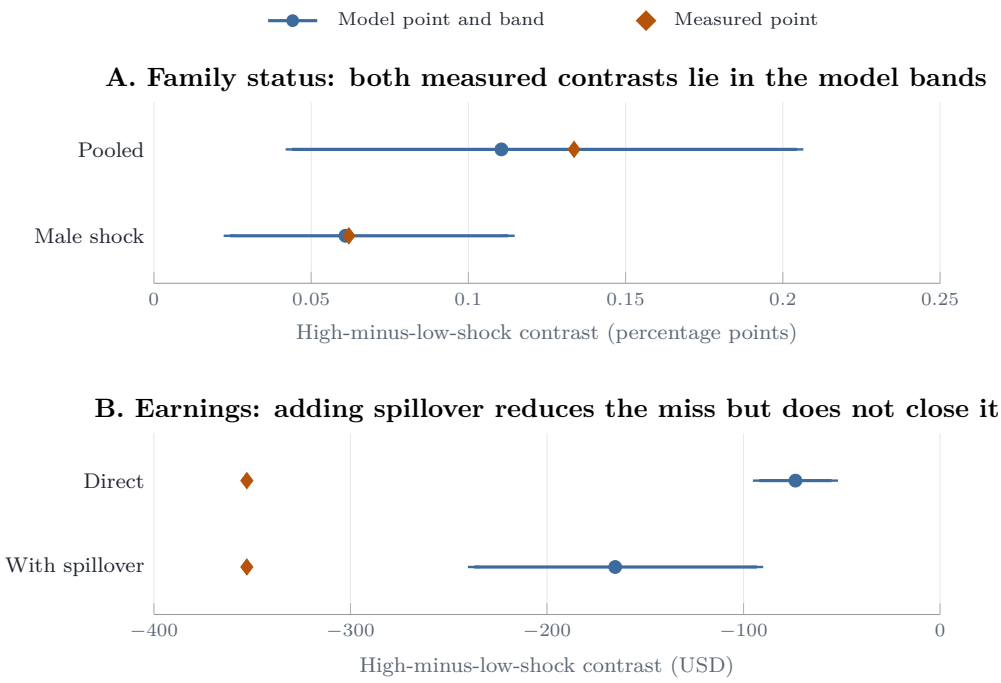


Figure 4: Within-period-demeaned commuting-zone contrasts from 1444 zone-period rows. Model points and bands come from the same generated scorecard as the tables. Measured diamonds omit sampling uncertainty; point coverage alone is not validation. The wage-spillover addition followed the original miss and is not an independent held-out success. Axes retain the outcomes' different units.

manufacture independent study counts. The source-selection and evaluation protocol in `docs/research/scenario_evidence_protocol.md` records a locally frozen research round, not retrospective preregistration or an independently timestamped prospective forecast. No new matched independent event score is established by these software and baseline-data checks.

6 Limitations

The model conditions on displacement and does not identify who was displaced or why. An upstream employer exposure can therefore be more uncertain than the downstream parameter band suggests. The entity adapter checks arithmetic and provenance fields, not the factual or causal validity of the supplied records.

Studies differ in country, age, sex, tenure, and measurement window. Transporting them across populations is a substantive assumption. Default mortality uses a constant baseline and the five-phase source-offset profile described above. The separate age-stratum projection advances age-band baselines under a different, constant source-window response assumption; neither models individual risk factors or resolves source-to-target transport. Lifetime earnings conversion uses a synthetic cross-sectional profile. Local employment multipliers omit migration and equilibrium adjustment. These choices can affect magnitudes even when arithmetic and citations are correct.

Transmission across distant generations is weakly identified. Parallel family pathways may overlap, place adjustments transfer a rank effect to an earnings scale, and repeated place steps can magnify the adjustment. None of these is resolved by collecting more Monte Carlo draws.

The current bands use heterogeneous evidence conventions and simplified marginal shapes. Only the declared correlation pairs enter Monte Carlo; other dependencies remain unmodeled. Place measurements and baseline estimates are generally fixed in default scenario sampling. The separate national uncertainty budget includes documented baseline-count and declared exposure distributions under additional assumptions.

Household intervals and county posterior sampling remain unavailable. Structural ensembles cover only selected alternatives. Internal coverage tests therefore validate machinery, while V1 exposes economic and measurement discrepancies. There is no basis for treating the output as a complete uncertainty interval for an individual, employer, or community.

The repository also records refused and contested links. It does not encode a positive immigration-to-crime effect from a pooled literature that does not support one [44, 11]. Observational findings may supply cross-checks without setting a causal point. Named competing specifications must remain visible rather than being averaged into a preferred headline.

7 Prioritized development

The next work should evaluate the experimental capabilities now implemented. The national baseline/exposure budget, age-stratum mortality projection, child present values, and household comparison each retain specific applicability and uncertainty limits. Extend baseline sampling to compatible county inputs, quantify target-population transport, and assess the constant-hazard conversion. Resolve the place counterfactual before adding further place-driven descendants. Review effect scale, intervention, mediator versus total effect, source population, and window metadata; compatible units alone do not establish causal composability.

Expand compatible independent study pools with overlap and window checks and small-sample synthesis sensitivity. Preserve null, protective, and conflicting effects. Broaden age/sex/tenure response support only with source estimates; adding baseline strata alone does not identify effect modifiers. Estimate survey-design uncertainty and assess longer-run stability of the SIPP transition cells. For the already separate manufacturing and high-tech job multipliers, test whether expansion effects transfer symmetrically to displacement.

Validation should prioritize one completed independent event before expanding the graph: freeze a matched exposure bridge, geography, population, outcome window, and observation model, then report every scored outcome. Register one prospective forecast with an independent timestamp. Numerical tests should include null-intervention invariance, same-place counterfactual cancellation, independently seeded sampling designs, and known analytic variance shares. These checks complement the existing property tests; passing code tests cannot repair an unidentified causal bridge.

7.1 A person-and-place research program

Figure 5 summarizes the implemented household prototype and the research gates that still precede public use.

The longer-term target is a distribution of possible outcomes conditional on a supplied profile and specified event, with an explicit reference scenario. It requires a baseline model of how employment, income, health, household relationships, and location evolve without that event. Baseline transitions may be estimated from documented longitudinal data; their predictive associations must be distinguished from identified causal intervention effects and from declared structural assumptions.

Personalization should distinguish characteristics that change baseline risks from those supported as effect modifiers. A location field does not establish a county-specific treatment effect. Linked household members, children's ages, shared resources, and event histories are necessary for family consequences. Community spillovers require an appropriate population model; summing person-level effects does not identify every regional response. Event definitions must include onset, duration, dose, comparison condition, and ordering. Unsupported interactions remain unresolved even when each event has an individual effect estimate.

The implemented five-year synthetic household demonstration compares employment probabilities and gross earnings with and without a supplied displacement duration. The next research step is to evaluate that restricted comparison against simple benchmarks and independently matched outcomes, then extend linked family dynamics only where justified. A protective intervention can follow when independently supported. Shared random draws can improve sampled comparison precision; the current household prototype instead calculates marginal expectations exactly. Neither identifies one person's joint potential outcomes. A full

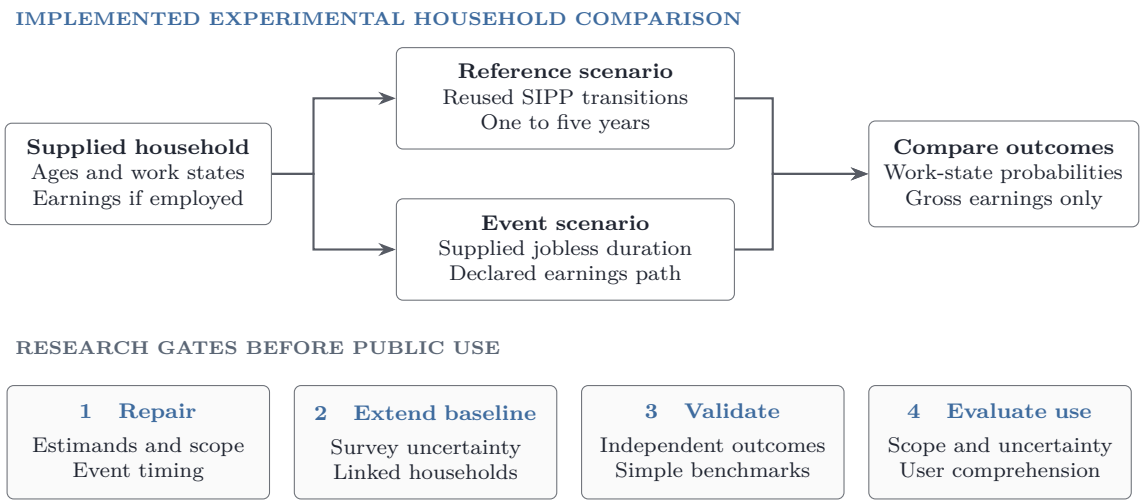


Figure 5: The implemented household prototype and its remaining release gates. Both scenarios start from the same supplied household. Observed transitions and supplied unemployment duration define conditional expectations, not an identified causal recovery path. Joint household dynamics, further events, community feedback, and longer horizons require additional evidence and validation.

life-course simulator and public applicability are not established. Milestones and acceptance criteria are recorded in `docs/research/life_course_research_plan.md`.

7.2 Public calculator as a research interface

An eventual website should make the profile, event, comparison, outcome unit, time horizon, and evidence applicability visible together. It should show absolute outcomes and scenario differences, distinguish uncertainty about the model from variation among simulated lives, and separate missing evidence from a supported null effect. Information should unfold from a short comparison to trajectories, assumptions, and study receipts, without requiring the visitor to understand a graph before using the calculator. Neutral scenario colors, legible charts, accessible controls, and clear unsupported-result states are part of scientific communication.

Initial public use should be restricted to evaluated scenarios and supported profiles. A synthetic example can demonstrate interaction before those capabilities are ready, but must be labeled as illustrative. Usability evaluation should measure whether visitors understand the comparison and its limits, rather than treating visual polish or expressed trust as evidence of comprehension. The separate product plan, `docs/research/website_experience_plan.md`, specifies the intended journey, visual direction, explanation patterns, and release criteria; it is a design proposal, not a description of a deployed service.

8 Reproducibility and availability

The model and paper source are provided under the repository's MIT license. Run the tests, `downstream audit`, and `downstream validate` to inspect implementation and evidence checks. Sampling commands record their seeds and draw counts. Parameter exports carry a version; the paper also records a content digest because parameter version alone does not identify engine code changes. The September 10 review repaired county predictive-baseline consistency, disconnected custom chains, ignored chain starting values, and the fixed copula seed used in sensitivity designs. Correlated Monte Carlo results can consequently change at the same historical user seed.

For an explicitly declared exposure basis, display scaling multiplies expected outcomes and their corresponding experiments by a nonnegative factor s; output variance scales by s^2. This is a display transformation, not a new cohort fit. Compatible place mixtures combine expectations as $\sum_k w_k E[X_k]$,

retaining every component's baseline and applicability. Linearity of expectation does not require independent components, but it also does not identify a joint distribution. Quantiles and variance indices cannot generally be averaged. A positive-weight unavailable component leaves the total unavailable rather than contributing zero. The consumer helpers therefore withhold joint mixture intervals until a joint draw/dependence calculation exists.

Running `make -C paper` regenerates tables and chart data, builds the PDF, and writes `paper/generated/results.json`. This JSON contains the full validation output and the source SHA-256. The appendix and coverage interpretation use the same live scorecard; reference mortality persistence values are generated from the scenario code. The archived follow-through evaluation is separately labeled in the JSON. The generated appendix preserves parameter points, bands, tiers, and citation keys. Population scope, distribution declarations, and extraction notes remain in the underlying CSV files. Source content digest for this build begins `10e687a538370d9d`. Generated numerical results are not manually copied into the prose. Vector diagrams and charts use the same restrained palette throughout. Model bands and measured points differ in both color and marker shape. Chart coordinates retain the engine's unrounded values; the tables provide compact rounded readings. The digest manifest includes engine code, parameter and validation CSV, JSON and XML inputs, and build scripts/configuration. A semantic HTML companion at `docs/methods.html` supplies readable text, scoped tables, baseline qualifications and a complete CSV download. The PDF remains an untagged publication artifact; the HTML companion is the accessible reading path.

A Numerical validation scorecards

These tables retain the exact comparison structure behind Figures 3 and 4. Values are rounded to four decimal places; chart coordinates use the unrounded engine output. "Covered" means only that the measured point lies in the model band, not that the model is causally validated.

Table 2: Mortality per 100,000 adults per unit shock. Coverage asks whether the measured point lies in the modeled band.

Variant	Model point [band]	Measured	Covered?
all adults	18.78 [5.51, 42.87]	4.27	no
all male	27.17 [7.98, 61.94]	4.27	no
all male age matched	25.88 [7.6, 59.02]	4.27	no

Table 3: Unit divorce-path predictions in percentage points of women, compared with the measured family-status stock change.

Window (years)	Model point [band]	Measured
5	0.0306 [0.0096, 0.0909]	0.28
10	0.0611 [0.0192, 0.1819]	0.28

Table 4: Within-period-demeaned panel contrasts. Coverage uses the measured contrast as a point target.

Outcome	Model point [band]	Measured	Covered?
Divorce, pooled (pp)	0.1105 [0.0439, 0.2045]	0.1336	yes
Divorce, male shock (pp)	0.0609 [0.0242, 0.1127]	0.062	yes
Earnings, direct (USD)	-73.59 [-91.99, -55.19]	-352.73	no
Earnings, with spillover (USD)	-165.31 [-237.03, -93.24]	-352.73	no

B Current parameter set

The table is an inventory, not a claim that every row belongs to one composable causal chain. Values use each destination node's declared unit in `params/nodes.csv`. Different rows cannot be compared as if their numbers shared a scale. Parameter values below are rounded to four decimal places; the source CSV retains full precision and identifiers. Readable relationship labels are used here to prevent identifier fragments. A semantic HTML companion is generated at `docs/methods.html`.

Table 5: Generated parameter inventory, v1.53.

Relationship	Point [band]	Tier	Sources
displacement → worker earnings	0.8 [0.75, 0.85]	EXACT-abstract	[32, 43, 23]
earnings shock → mortality sustained	1.1354 [1.0335, 1.2474]	canonical	[53]
earnings shock → mortality peak	2.6698 [2.1144, 3.3711]	canonical	[53]
earnings shock → mortality offset 1	1.9838 [1.3697, 2.8732]	canonical	[53]
earnings shock → mortality offsets 2 3	1.7315 [1.3238, 2.2649]	canonical	[53]
earnings shock → mortality offsets 4 5	1.2712 [0.97, 1.6661]	canonical	[53]
displacement → divorce hazard	1.11 [1.05, 1.25]	canonical	[45, 14]
displacement → child earnings	0.9076 [0.844, 0.976]	EXACT	[43]
child earnings → grandchild earnings	0.55 [0.4, 0.6]	canonical	[51, 20, 16]
displacement → local service jobs	5 [1.6, 5]	canonical	[42]
school spending → child earnings	1.07 [1.03, 1.1]	canonical	[31]
youth wages → youth crime	-1 [-1.5, -0.5]	canonical	[25, 26]
displacement → infant birth weight	0.9541 [0.912, 0.9981]	EXACT	[37]
import shock → non displaced wage spillover	-0.822 [-1.3042, -0.3398]	EXACT	[3]
family income shock → child achievement sd	0.061 [0.0157, 0.1063]	EXACT	[21]
family income shock → life expectancy years	0.1333 [0.1167, 0.15]	EXACT	[17]
male earnings → marital fertility	1.24 [0.397, 2.083]	EXACT	[33]
male job loss rate → child maltreatment	6 [3.65, 8.35]	EXACT	[38]
displacement → grade retention hazard	1.1473 [1.0119, 1.2827]	EXACT	[52]
eviction order → emergency shelter use	4.778 [1.076, 8.48]	EXACT	[19]
foreclosure order → house price gap	0.99 [0.9875, 0.9925]	EXACT-results	[12]
local unemp shock → household ipv	1.58 [1.49, 1.58]	EXACT-results	[48]
household hardship → household ipv	2.14 [1.61, 2.68]	EXACT-results	[48]
couple unemployment → household ipv	1.3 [1.04, 1.63]	EXACT-results	[48]
neighborhood exposure → child outcomes modifier	0.037 [0.031, 0.043]	EXACT	[15]
unconditional income → child education years	1.127 [0.247, 2.007]	EXACT	[1]
unconditional income → youth any crime	-0.224 [-0.377, -0.071]	EXACT	[1]
unconditional income → youth minor crime ever	-0.179 [-0.353, -0.005]	EXACT	[1]
youth crime conviction share → youth crime convicted	0.061 [0, 0.123]	EXACT	[22]
youth violent crime conviction share → youth crime convicted	0.366 [0.143, 0.589]	EXACT	[22]
displacement → circulatory mortality hazard	1.54 [1.29, 1.83]	canonical	[10]
displacement → alcohol mortality hazard	1.62 [1.09, 2.41]	canonical	[10]
displacement → external cause mortality hazard	1.53 [1.15, 2.04]	canonical	[10]
displacement event → spouse mental health sd	-0.194 [-0.327, -0.061]	EXACT	[39]
displacement event → child achievement sd	-0.0206 [-0.0402, -0.001]	EXACT	[5]
displacement event → hs completion	-0.037 [-0.0801, 0.0061]	EXACT	[8]
displacement event → college attendance	-0.063 [-0.1081, -0.0179]	EXACT	[8]
displacement event → college completion	-0.036 [-0.0674, -0.0046]	EXACT	[8]
displacement event → adult depression cesd	0.025 [0.0034, 0.0466]	EXACT	[8]
displacement event → spouse participation elasticity	-0.04 [-0.07, -0.03]	EXACT	[27]
displacement event → annual birth response	-0.005 [-0.0089, -0.0011]	EXACT	[29]
displacement event → child depression anxiety	0.008 [0.0002, 0.0158]	EXACT	[47]
child achievement sd → grandchild achievement sd	0.38 [0.38, 0.42]	EXACT-results	[7]
child education years → grandchild education years	0.296 [0.255, 0.337]	EXACT-results	[36]
infant birth weight → child earnings	0.1 [0.0216, 0.1784]	EXACT	[6]
infant birth weight → adult type2 diabetes hazard	-0.9637 [-1.3834, -0.5401]	EXACT	[34]

Relationship	Point [band]	Tier	Sources
infant birth weight → adult cardiovascular disease hazard	-0.6994 [-0.8173, -0.585]	EXACT	[34]
divorce hazard → child divorce hazard	1.88 [1.57, 2.24]	EXACT-results	[41]
married cohort → parental dissolution share	0.069 [0.069, 0.072]	EXACT-results	[45]
undocumented share change pp → receiving violent crime log rate	-0.12 [-0.18, -0.06]	EXACT	[35]
regular immigrant share ratio → receiving total crime rate	0.105 [-0.2616, 0.4716]	canonical	[4]

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